

YEAR 12 CHEMISTRY - 3ACHE



TEST 1

Atomic Structure, Bonding and Solutions

Recommended time: 1 hour

Total marks: 40

Name:

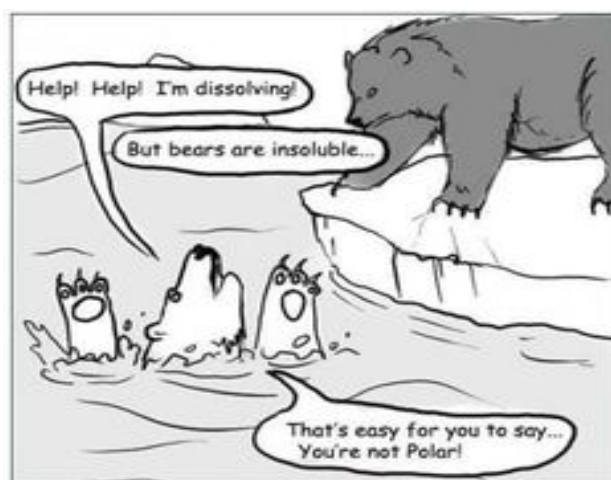


Teacher:

This test is in two parts.

Part 1: Multiple choice style test consisting of **TEN (10)** questions.
Each question is worth 1 mark.
Write your answers in the **table** provided on this page.
Attempt ALL Questions

Part 2: Short and/or Extended Answer questions
Write all answers in the spaces provided.
The marks allocated to each question are shown at the foot of each question.



PART 1: Multiple Choice

Place your answer in the table below. This part is worth 10 marks.

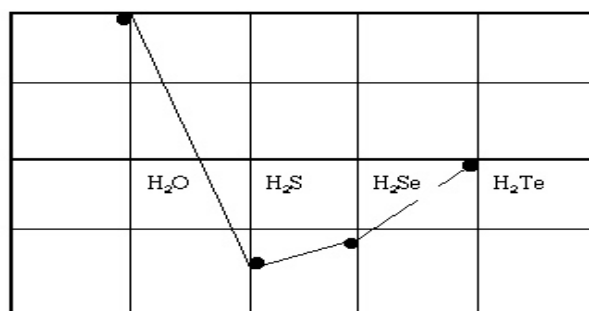
1	☒ a	☐ b	☐ c	☐ d	6	☐ a	☐ b	☐ c	☒ d
2	☐ a	☐ b	☒ c	☐ d	7	☐ a	☒ b	☐ c	☐ d
3	☐ a	☒ b	☐ c	☐ d	8	☐ a	☐ b	☒ c	☐ d
4	☒ a	☐ b	☐ c	☐ d	9	☐ a	☒ b	☐ c	☐ d
5	☐ a	☐ b	☒ c	☐ d	10	☒ a	☐ b	☐ c	☐ d

PART 2: Short Answers**30 marks**Answer **all** questions. Write your answers in the spaces provided.**Question 1**(a) **Na⁺ or Mg²⁺ or Al³⁺ or Si⁴⁺ etc** **1 mark**(b) **2 8** **1 mark****Question 2**

Complete the table below. For Lewis structures, any lone pairs must be shown.

All electron shell pairs should be represented as either : or as —.

Species	Lewis structure (show all valence electrons)	Sketch or name of molecular shape	Polarity of molecule (polar or non-polar)
NF ₃		Pyramidal or 	Polar
SO ₂		Bent or 	Polar
HCN		Linear or 	Polar

1 mark per box (total = 9 marks)**Question 3**Using the axes provided, sketch a **line graph** to show the trend observed in the boiling points of Group 16 hydrides H₂O, H₂S, H₂Se and H₂Te.**Boiling point****1 mark****1/2 mark if not showing anomaly of trend from H₂O**

Question 4

By referring to the structure and/or bonding present, **explain** why sulfur (2.6) is not as electronegative as nitrogen (3.0) even though it has more protons. (3 marks)

S atom has one more electron shell 1 mark

Shielding effect due to repulsion from inner shell electrons in S is higher than P 1 mark

Nuclear pull in S is thus smaller resulting in a decrease in ability to attract bonding electrons 1 mark

Question 5

By referring to the structure and/or bonding present, **explain** why the melting points of Group 1 elements decrease down the group but the melting points of Group 17 elements increase down the group. (5 marks)

Group 1:

. has more electron shells (or larger radius) going down the group 1 mark

. shielding effect due to repulsion from inner shell electrons increases down the group 1/2 mark

. less energy is required to overcome the weaker attraction between positive nuclei and delocalized electrons (or metallic bond) i.e. a decrease in melting points 1 mark

Group 17:

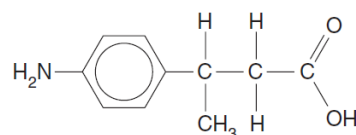
. more electrons going down the group 1/2 mark

. frequency of temporary dipoles increases due to greater volume of space 1 mark

. more energy is required to overcome the stronger dispersion forces between molecules i.e. an increase in melting points 1 mark

Question 6

- (a) Many drugs are produced by chemically combining several molecules. One example is an amino acid that, in combination with another substance, has the potential to be a new anti-inflammatory drug. The structure of this amino is shown below:



Use the structure to determine if this substance is likely to be soluble or insoluble in water. Briefly **explain** your choice using your knowledge of bonding and solubility. (3 marks)

. Soluble in water 1 mark

. Present of electronegative N and O atoms 1 mark

. Hydrogen bonds can be formed between the electronegative N and O atoms with the partially positive H atoms of water molecules (and/or between the H atoms in the – NH₂ and – OH groups and the partially negative O atoms of water molecules) 1 mark

- (b) Barium iodide (BaI₂) is soluble in water. Circle ONE (1) interaction from the list below that **best** explains this behaviour.

VAN DER WAALS

DIPOLE-DIPOLE

COVALENT BONDS

DISPERSION FORCES

COVALENT-DIPOLE

ION-DIPOLE

HYDROGEN BONDS

IONIC BONDS

1 mark

Question 7

A 5.0 g sample of chlorine gas (Cl₂) was fully dissolved in 100.0 L of water at room temperature. Calculate

- (a) the concentration of chlorine in mol L⁻¹. (2 marks)

$$n(\text{Cl}_2) = 5 / 2(35.45) = 0.07052 \text{ mol} \quad 1 \text{ mark}$$

$$c(\text{Cl}_2) = 0.07052 / 100 = 7.052 \times 10^{-4} \text{ mol L}^{-1} \quad 1 \text{ mark}$$

- (b) the concentration of chlorine in ppm if 1.0 mL of water has a mass of 1.0 g. (2 marks)

$$m(\text{Cl}_2) \text{ solute} = 5000 \text{ mg} \quad 1/2 \text{ mark}$$

$$m(\text{solution}) = (100 + 5 \times 10^{-3}) \text{ kg} \quad 1/2 \text{ mark}$$

$$\text{ppm} = 5000 / (100 + 5 \times 10^{-3}) = 50 \text{ ppm (49.999 to be exact!)} \quad 1 \text{ mark}$$

$$(\text{or } 50 / 100,005 \times 10^6)$$

Question 8

The label on the side of a 700 mL bottle of whisky is shown below

37% alc/vol

CONTAINS APPROX 20 STANDARD DRINKS

Note: 37% alc/vol means that every 100 mL of wine contains 37 mL of pure ethanol ($\text{C}_2\text{H}_5\text{OH}$). The density of pure ethanol is 0.790 g mL^{-1} at room temperature.

Use the information provided to calculate the mass of pure ethanol in a 700 mL bottle at room temperature.

(2 marks)

V (pure ethanol) in 700 mL of wine = 37×7 = 259 mL 1 mark

m (pure ethanol) = 259×0.79 = 204.6 g 1 mark

END OF TEST